# Uncertainty intervals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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## Prepare the purpose, prerequisites and materials

> Prepare Uncertainty intervals: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of uncertainty intervals

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Sampling variability

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sampling variability

Reference example, not the only acceptable wording: Teacher explanation: Sampling variability means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

Consider finite population values1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size2 without replacement. Their means are1, 2, 2, 2, 2, 3.

### Experimental units and independence

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of experimental units and independence

Reference example, not the only acceptable wording: Teacher explanation: The experimental unit is the independently assigned or sampled entity. Multiple measurements from one unit are often related.

If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.

### A literature comparison matrix

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of a literature comparison matrix

Reference example, not the only acceptable wording: Teacher explanation: A literature matrix compares studies using aligned columns rather than sorting headline numbers with different meanings.

Record question, data, unit, baseline, metric, split, resources and limitations. Mark incomparable fields rather than forcing a ranking.

Two 45-minute sessions. An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the **mechanism** and terminology of uncertainty intervals

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

### Prerequisite answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

### End goal for the pictorial case

Explain uncertainty intervals using the records and boundaries of this fictional investigation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain and model the mechanism

> Model Uncertainty intervals using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

### Explain Uncertainty intervals · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For this constructed exercise assume independent normal measurements with known population standard deviation2. A95% mean interval uses supplied normal critical value1.96:mean plus/minus1.96*sigma/sqrt(n).

Un**known sigma** requires a different justified method.

## Explain Uncertainty intervals · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

| Term | Meaning |
| --- | --- |
| Standard error | The model-based variation of an estimate |
| Confidence interval | An interval procedure with declared repeated-sampling coverage |
| Known sigma | A supplied population standard deviation assumption |

### Uncertainty intervals — evidence trace

**INPUT AND SCOPE**

Observed mean10, n16, sigma2. Compute standard error, margin and interval.

**CHECK OR CALCULATION**

For this constructed exercise assume independent normal measurements with known population standard deviation2. A95% mean interval uses supplied normal critical value1.96:mean plus/minus1.96*sigma/sqrt(n). Unknown sigma requires a different justified method.

**CONCLUSION AND LIMIT**

SE=2/4=.5; margin=.98; interval[9.02, 10.98]. Under assumptions, the method covers the fixed population mean in95% of repeated samples.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

## Explain Uncertainty intervals · 3

Guided: SE=2/4=.5; margin=.98; interval[9.02, 10.98]. Under assumptions, the method covers the fixed population mean in95% of repeated samples.

Qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Observed mean10, n16, sigma2. Compute **standard error**, margin and interval.

SE=2/4=.5; margin=.98; interval[9.02, 10.98]. Under assumptions, the method covers the fixed population mean in95% of repeated samples.

## Model the reasoning aloud · 4

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

## Model the reasoning aloud · 5

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from math import sqrt

def known_sigma_interval(mean, sigma, n, critical=1.96):
    """Assume independent normal sampling and a known population sigma."""
    if type(n) is not int or n < 1 or sigma < 0 or (critical <= 0):
        raise ValueError('interval settings')
    se = sigma / sqrt(n)
    margin = critical * se
    return (se, mean - margin, mean + margin)

print(known_sigma_interval(10, 2, 16))
```

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Uncertainty intervals

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

### The facts we will use

> **Supplied facts**
>
> Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

AI ACADEMY / AI-7/12

# Supplied interval for B minus A time

| | Lower endpoint (s) | Estimate (s) | Upper endpoint (s) |
|---|---|---|---|
| Original supplied interval | -1.4 | -0.8 | -0.2 |
| Changed supplied interval | -1.4 | not specified | 0.3 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/12

# Uncertainty intervals

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Locate the estimate

-0.8 lies between the interval endpoints.



## 2. Interpret the sign

Negative B-A means B took less time under this measurement convention.



## 3. Respect coverage meaning

95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

**CHANGE ONE THING**

Use a supplied interval [-1.4,0.3] instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Locate the estimate | -0.8 lies between the interval endpoints. |
| 2. Interpret the sign | Negative B–A means B took less time under this measurement convention. |
| 3. Respect coverage meaning | 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

### What this example does not establish

#### Keep the stated limit

An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

### Change one thing in the pictured case

#### Change one thing

Change: Use a supplied interval [-1.4,0.3] instead.

#### Predicted result

**Prediction**: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval.

#### Why the result follows

Reason: The new upper endpoint crosses zero.

### Ask while showing the picture

#### Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Uncertainty intervals, reduce support deliberately and check individual reasoning before discussing answers.

| Time or stage | Teacher action |
|---|---|
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation. |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

### Reduce help deliberately

### Completion practice

Completion task: Observed mean10, n16, sigma2. Compute **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

### Reasoning to listen for

Teacher check: Teacher explanation: SE=2/4=.5; margin=.98; interval[9.02, 10.98]. Under assumptions, the method covers the fixed population mean in95% of repeated samples.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

#### Individual reasoning check

Use the worked case for Uncertainty intervals. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Expected repair

Expected correction: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

#### Pictorial practice question

Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

**Reasoned teacher answer**

-0.8 lies between the interval endpoints. Negative B–A means B took less time under this measurement convention. 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Changed case: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Changed-case reasoning: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the investigation and diagnose misconceptions

> Run the Uncertainty intervals investigation and use the first unsupported step to choose a teaching response.

The worked result establishes every future case: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Uncertainty intervals questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Guided and exit questions

Guided: SE=2/4=.5; margin=.98; interval[9.02, 10.98]. Under assumptions, the method covers the fixed population mean in95% of repeated samples.

Qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### W1 Explain the mechanism

#### Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

#### Teacher reasoning · W1

Teacher explanation: An uncertainty interval needs a sampling model and assumptions. A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

For this constructed exercise assume independent normal measurements with known population standard deviation2. A95% mean interval uses supplied normal critical value1.96:mean plus/minus1.96*sigma/sqrt(n).

Un**known sigma** requires a different justified method.

### W2 Trace the worked case

#### Workbook question · W2

Observed mean10, n16, sigma2. Compute **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

#### Teacher reasoning · W2

Teacher explanation: SE=2/4=.5; margin=.98; interval[9.02, 10.98]. Under assumptions, the method covers the fixed population mean in95% of repeated samples.

### W3 Audit the limit

#### Workbook question · W3

A report omits this qualification: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

#### Teacher reasoning · W3

Teacher explanation: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

### W4 Apply a new case independently

#### Workbook question · W4

Mean20, n25, **known sigma**5 and critical1.96. Compute interval and name one assumption.

#### Teacher reasoning · W4

Teacher explanation: SE1, margin1.96, interval[18.04, 21.96]. Require independent normal sampling and **known sigma**; bias is not repaired by the interval.

### W5 Design a revealing follow-up

#### Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

#### Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

The new case should test this particular **mechanism** rather than merely restate the original result.

### Independent assessment and fresh retry

#### Teacher key · Independent assessment and fresh retry

W4: SE1, margin1.96, interval[18.04, 21.96]. Require independent normal sampling and **known sigma**; bias is not repaired by the interval. Require intermediate reasoning and the stated scope.

Fresh retry: Mean8, n16, **known sigma**4. Expected reasoning: SE1, margin1.96, interval[6.04, 9.96] under the same assumptions.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: Mean20, n25, **known sigma**5 and critical1.96. Compute interval and name one assumption.

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: SE1, margin1.96, interval[18.04, 21.96]. Require independent normal sampling and **known sigma**; bias is not repaired by the interval.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect the project, support and next lesson

> Connect Uncertainty intervals to Reproducible AI Research Study, adapt support and plan a later changed-case check.

Add a labelled uncertainty intervals **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of uncertainty intervals.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Effect size and practical importance. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

### Plan the delayed return

Suggested return points in this level: ai-7/13, ai-7/15, ai-7/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Uncertainty intervals lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Explain the **mechanism** and terminology of uncertainty intervals

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Observed mean10, n16, sigma2. Compute **standard error**, margin and interval. Show the intermediate steps and give the conclusion with its scope.

### Supported practice answer

Reasoned practice answer: Teacher explanation: SE=2/4=.5; margin=.98; interval[9.02, 10.98]. Under assumptions, the method covers the fixed population mean in95% of repeated samples.

Check every learner before discussion: Use the worked case for Uncertainty intervals. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: A narrow interval can be precisely biased; sampling independence and representativeness must be justified.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Use a supplied interval [-1.4,0.3] instead. Require a **prediction** and a reason before comparing.

Expected result: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. Reason: The new upper endpoint crosses zero.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
| --- | --- | --- | --- |
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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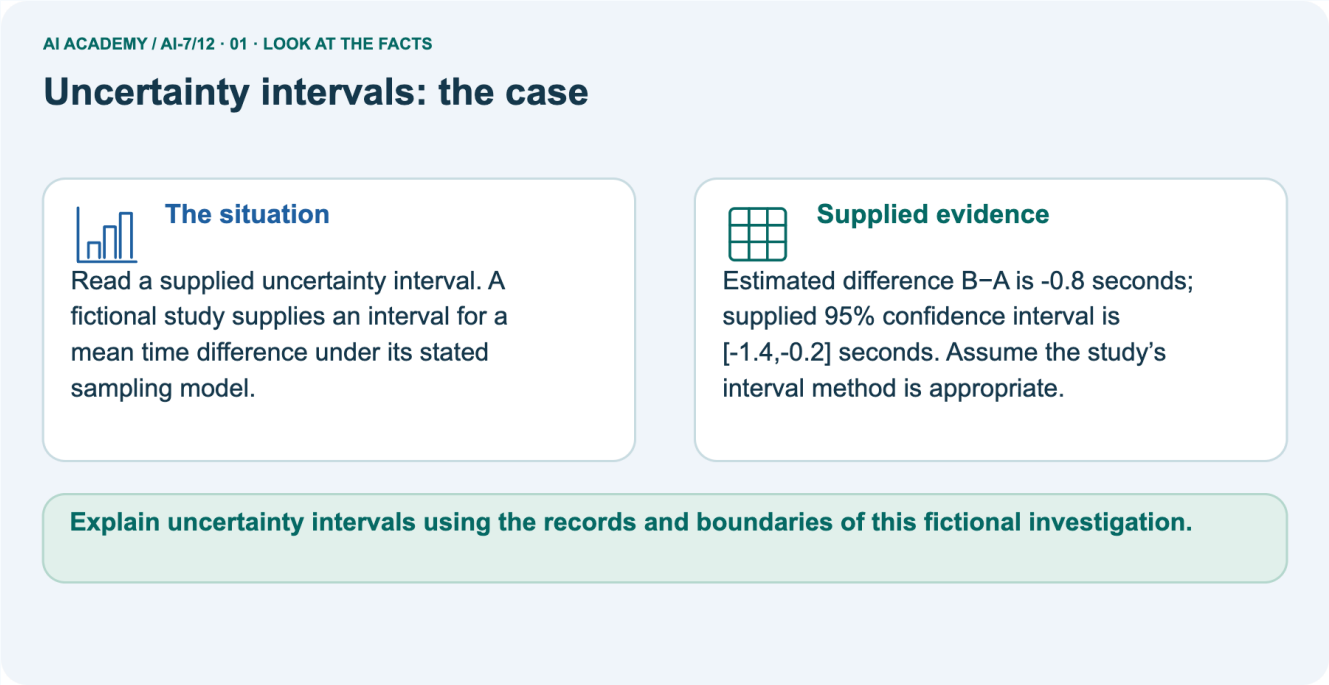
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## Visual concept journey

> Show one Uncertainty intervals graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Uncertainty intervals: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

Read a supplied uncertainty interval. A fictional study supplies an interval for a mean time difference under its stated sampling model.

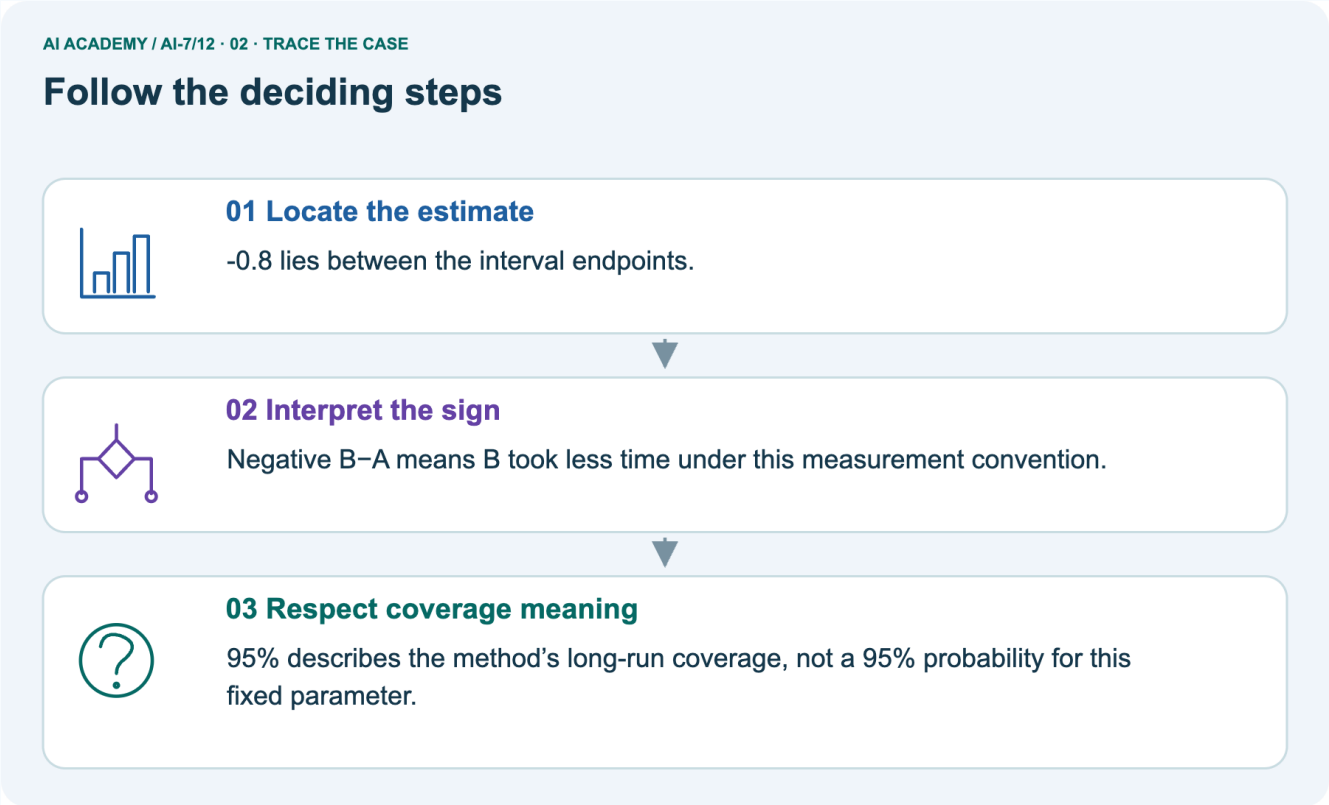
Supplied facts: Estimated difference B–A is -0.8 seconds; supplied 95% **confidence interval** is [-1.4,-0.2] seconds. Assume the study's interval method is appropriate.

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

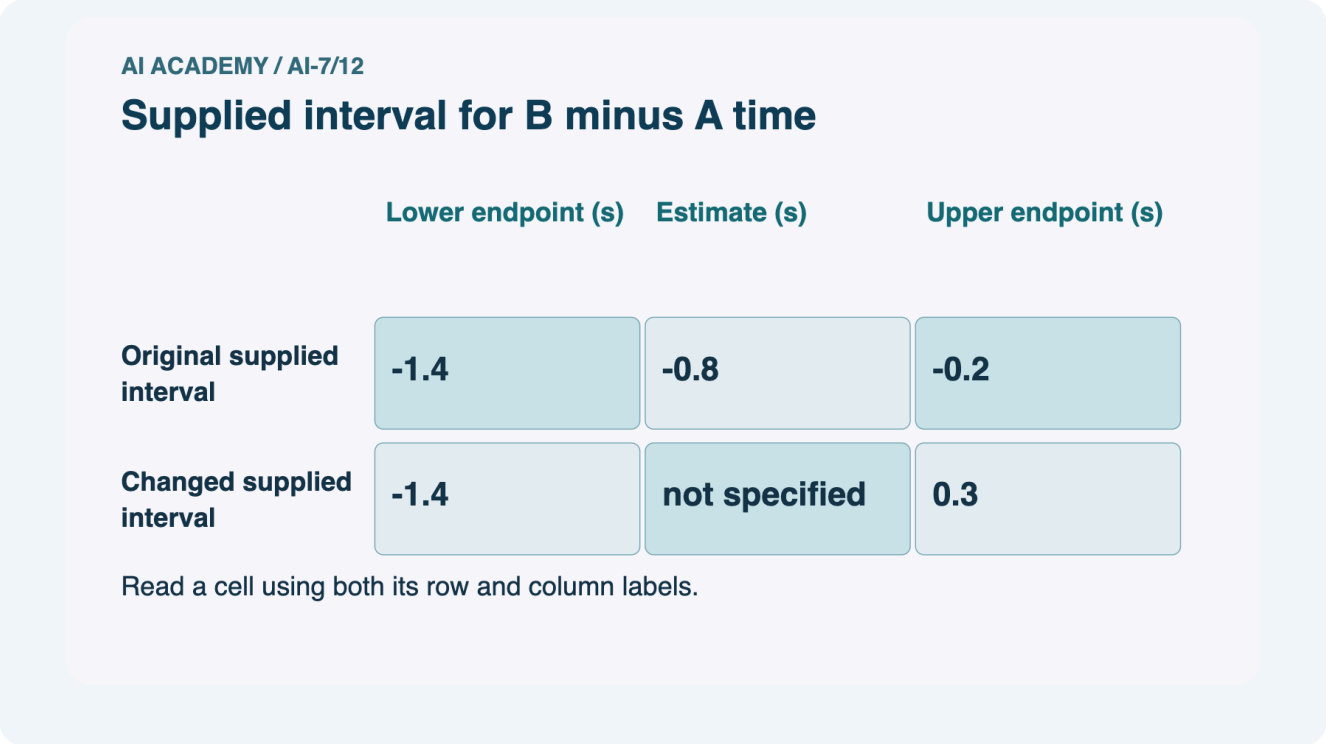
Locate the estimate: -0.8 lies between the interval endpoints. → Interpret the sign: Negative B–A means B took less time under this measurement convention. → Respect coverage meaning: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Supplied interval for B minus A time




Follow the labelled picture with the matching explanation. Open the vector diagram

Supplied interval for B minus A time. Original supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = -0.8, Upper endpoint (s) = -0.2.

Changed supplied interval: Lower endpoint (s) = -1.4, Estimate (s) = not specified, Upper endpoint (s) = 0.3.

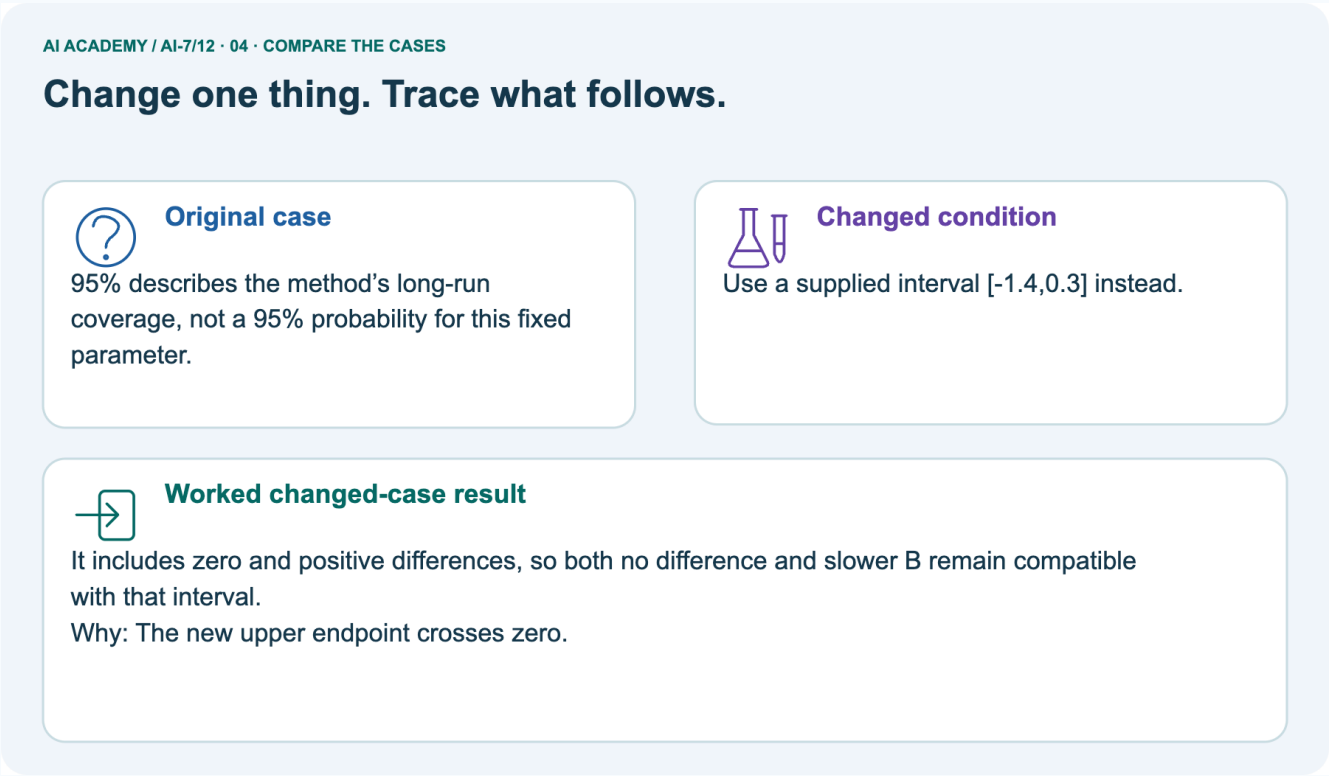
Read the labels and values. Explain the relationship the model shows.

#### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

### 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 95% describes the method's long-run coverage, not a 95% probability for this fixed parameter. Changed condition: Use a supplied interval [-1.4,0.3] instead.

Expected result: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. Reason: The new upper endpoint crosses zero.

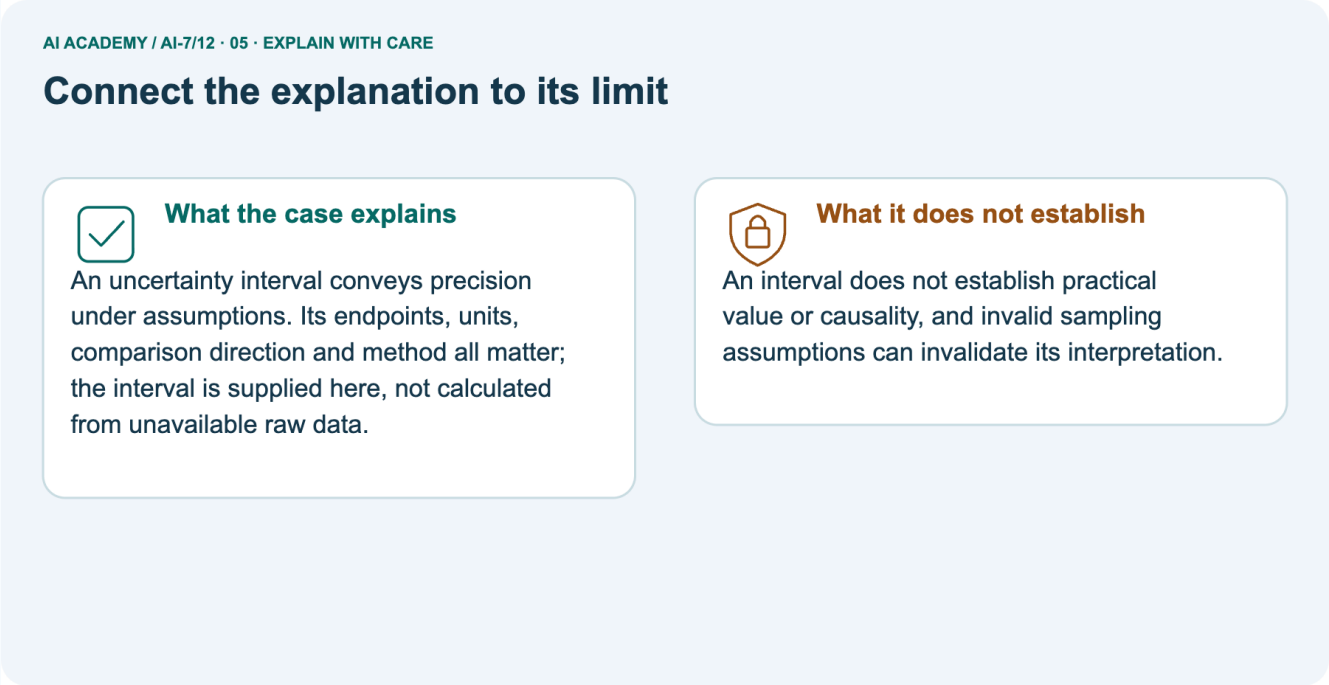
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The new upper endpoint crosses zero.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

An uncertainty interval conveys precision under assumptions. Its endpoints, units, comparison direction and method all matter; the interval is supplied here, not calculated from unavailable raw data.

Limit: An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

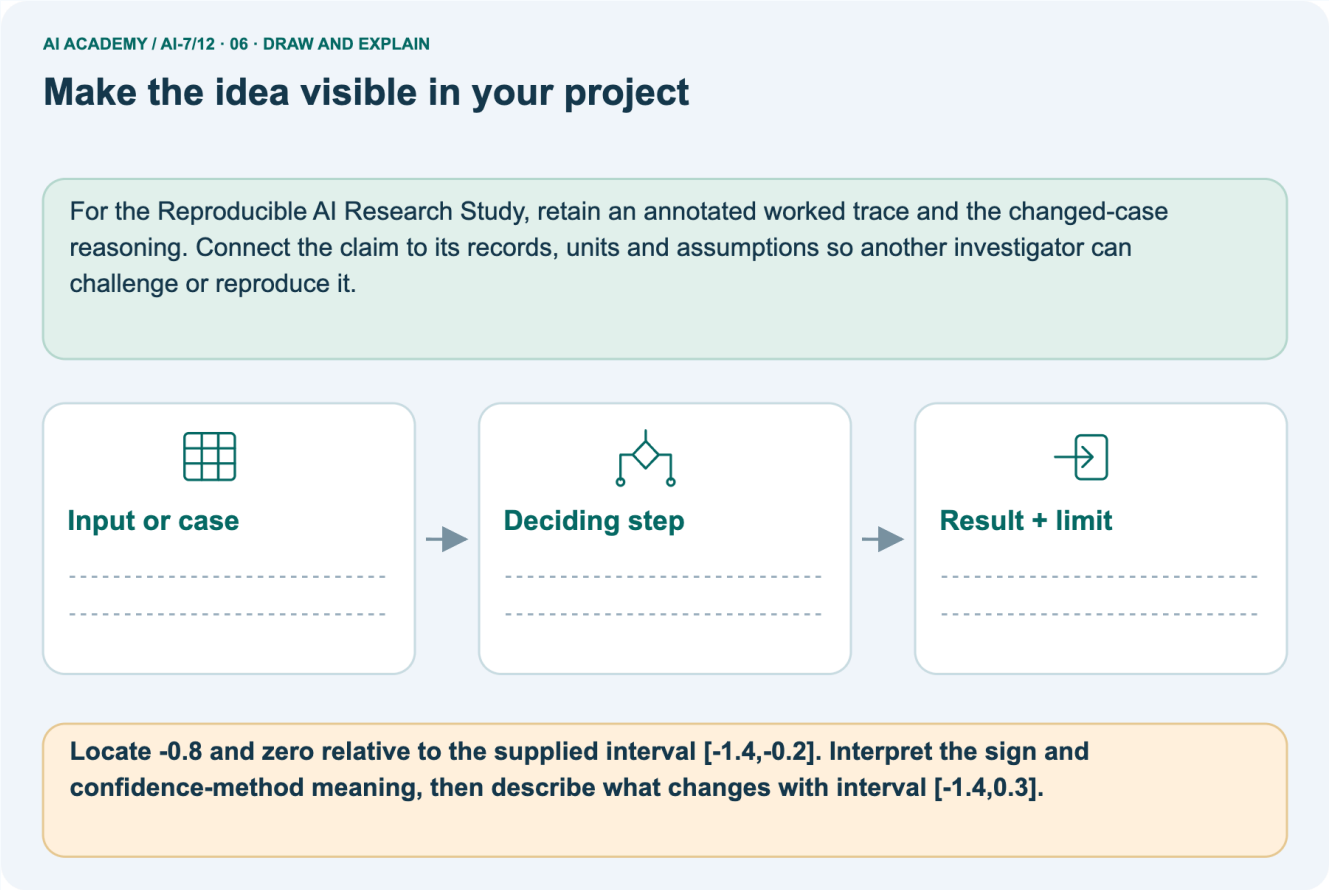
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: An interval does not establish practical value or causality, and invalid sampling assumptions can invalidate its interpretation.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: -0.8 lies between the interval endpoints.

Negative B–A means B took less time under this measurement convention. 95% describes the method’s long-run coverage, not a 95% probability for this fixed parameter.

Changed case: It includes zero and positive differences, so both no difference and slower B remain compatible with that interval. The new upper endpoint crosses zero.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.